

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) **Date: 22-02-26**
Time: 3 Hrs **Max. Marks: 180**

22-02-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE ADV_RTA-3_SYLLABUS

MATHEMATICS: **PRESENT WEEK (100%):** P and C, Probability/ Random variable, Vectors-3D

PHYSICS: **PRESENT WEEK (100%):** Electrostatics, Capacitors + RC Circuits, Gravitation, Magnetic effects

CHEMISTRY: **PRESENT WEEK (100%):** Surface Chemistry, Coordination compounds, Aldehyde & ketone

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TIME: 3HRS **Important Instructions** **Max Marks: 180**

MATHEMATICS :

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 8)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec – II (Q.N : 9 – 14)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 15 – 18)	Matrix Matching Type	+3	-1	4	12
Total				**18**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 26)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec – II (Q.N : 27 – 32)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 33 – 36)	Matrix Matching Type	+3	-1	4	12
Total				**18**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 44)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec – II (Q.N : 45 – 50)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 51 – 54)	Matrix Matching Type	+3	-1	4	12
Total				**18**	**60**

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MATHEMATICS **Max. Marks: 60**

SECTION - I
(Maximum Marks : 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

1. The shortest distance between $(1-x)^2+(x-y)^2+(y-z)^2+z^2=\frac{1}{4}$ and $4x+2y+4z+7=0$ in 3-dimensional coordinates system is equal to _____.

2. Let $\vec{r}$ be position vector of a variable point in xy-plane such that $\vec{r}\cdot\left(10\hat{j}-8\hat{i}-\vec{r}\right)=40$ and M = max. $\left\{\left|\vec{r}+2\hat{i}-3\hat{j}\right|^2\right\}$ and m = min. $\left\{\left|\vec{r}+2\hat{i}-3\hat{j}\right|^2\right\}$

   Also let $\vec{a}, \vec{b}, \vec{c}$ be vectors representing three co-terminus edges of a tetrahedron such that $\vec{a}\wedge\vec{b}=\vec{b}\wedge\vec{c}=\vec{c}\wedge\vec{a}=\frac{\pi}{3}$ and $4(\vec{a}\cdot\vec{a})+3(\vec{b}\cdot\vec{b})+2(\vec{c}\cdot\vec{c})=8(M+m)$. If V is the maximum volume of the tetrahedron, then the value of (2V + 5) is

3. The total number of numbers of the form $a_1a_2a_3a_4a_5$ where $a_1\le a_2\le a_3\le a_4\le a_5$ or $a_1\ge a_2\ge a_3\ge a_4\ge a_5$ and $a_1, a_2, a_3, a_4, a_5 \in \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is

4. Given 5 lines segments of length 2, 3, 4, 5, 6 and 7 units, then number of triangles that can be formed by joining these line segments is/are

5. Let X = {1, 2, 3, ….,90} and Y is a subset of X such that the sum of no two elements in Y is divisible by 7, then the maximum possible number of elements in Y is _____.

6. Three students A, B, C appear for IIT-JEE exam. Their respective probabilities of clearing the exams being $\frac{3}{10}, \frac{1}{2}$ and $\frac{2}{5}$. Each student clearing or not clearing the exam is independent of other. If the probability of A passing the exam be $\frac{m}{n}$ (m, n ∈ N) given that only one of the student has cleared the exam, then the least value of (m + n) is equal to

7. Mr. X walk up 16 steps, going up either 1 or 2 steps at a time. There is an explosive material on the 8th step so he cannot step there. The number of ways in which Mr. X can go up 16 steps is _____.

8. Let fair coin be tossed 6 times A, B and C three events are defined as

   A: exactly 4 heads are obtained.

   B: 4th head obtained on 5th toss.

   C: tail is obtained, on 2nd or 4th toss

   If conditional probability, $P\left(\frac{B}{A\cap C}\right)=\frac{m}{n}$ where m, n ∈ N then find the least value of (n–m).

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SECTION – II
(Maximum Marks : 24)

This section contains **SIX(06)** multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE THAN ONE** option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

9. In the tournament the player starts with i points and the computer starts with n – i points, where n, i are whole numbers. If the player loses a game, he gives 1 points to the computer and if he wins the game, the computer gives him one point. In order to win the tournament the player must end up with all the n points. The probability of winning a game (for the player) is $\frac{1}{3}$. All the games in a tournament are independent events. Which of the following is(are) correct. (Assume that no game end in draw)

(A) If P_i denotes the probability that the player wins the tournament when he starts with i points and the computer with n – i points and $P_i = a\,P_{i+1} + b\,P_{i-1}$, then the value of (a + b) is 1

(B) If probability that the players will win the Tournament when n = 6 and i = 2 is p, then 21p is equal to 1

(C) If P_i denotes the probability that the player wins the tournament when he starts with i points and the computer with n – i points and $P_i = a\,P_{i+1} + b\,P_{i-1}$, then the value of (a + b) is 2

(D) If probability that the players will win the Tournament when n = 6 and i = 2 is p, then 21p is equal to 2

10. Let OL, OM, ON be coterminous edges of a cuboid. OL = 1 cm, OM = 2 cm, ON = 3 cm. Let ℓ be the shortest distance between OL and skew body diagonal, m be the shortest distance between OM and skew body diagonal and n be the shortest distance between ON and skew body diagonal. Then the value of $\sqrt{13\ell^2 + 10m^2 + 5n^2}$ is less than

(A) 3 (B) 7 (C) 9 (D) 11

11. Let $\vec{a}\times\vec{r} + (\vec{r}\cdot\vec{b})\vec{b} = \vec{a}$ and $\frac{|\vec{a}||\vec{b}|}{2} \le \vec{a}\cdot\vec{b} \le \frac{|\vec{a}||\vec{b}|}{\sqrt{2}}$. If $f(\theta) = [\vec{r}\ \vec{a}\ \vec{b}]$ where θ is angle between vectors $\vec{a}$ and $\vec{b}$ (where $\vec{a}$ and $\vec{b}$ are unit vectors)

(A) Maximum value of f(θ) is $\frac{1}{\sqrt{2}}$ (B) Maximum value of f(θ) is $\frac{3}{2}$

(C) Minimum value of f(θ) is $\frac{1}{\sqrt{2}}$ (D) Minimum value of f(θ) is $\frac{1}{2}$

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12. Consider a matrix $M=\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix}$ where $a_i < a_{i+1}\ \forall\ i = 1, 2, \ldots, 8$ and a_i's are the divisors of a smallest natural number N. If P = Absolute value of det(M); Q = Value of Tr(M); $R = \vec{V}_1 \cdot \vec{V}_2$ (where $\vec{V}_1 = a_1\hat{i} + a_9\hat{j} + a_3\hat{k}$ and $\vec{V}_2 = a_4\hat{i} + a_2\hat{j} + a_7\hat{k}$) S = determinant of matrix adj(M). Then choose the correct choice(s)

(where det(M) is the value of determinant of matrix M and adj(M) is the adjoint of matrix M and Tr(M) denotes trace of matrix M)

(A) S – 2R + P = 118

(B) R – P – Q = 51

(C) 2P + 5Q + R – S = 39

(D) P + Q + 3R – S = 53

13. Let X be the number of ways to paint faces of a regular octahedron (it has eight equilateral triangular faces, 6 vertices, 12 sides) with 8 different colors is N (All colors to be used), then

(A) X is equal to the indistinguishable number of ways to name the vertices of a cube with letters A, B, C, D, E, F, G, H

(B) Sum of digits of X is 15

(C) Total number of positive divisors of X is 32

(D) X has exactly four prime factors

14. Two players A and B play a series of independent matches. Whoever first win two matches (not necessarily consecutive) wins the game. The probability that A wins, draws or loses a match are $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$ respectively. Which of the following is/are true

(A) Probability of A winning the game at the end of 11th match is

$${}^{10}C_1\left(\frac{1}{2}\right)^2\left(\frac{1}{3}\right)^9 + {}^{10}C_2\left(\frac{1}{2}\right)^2\left(\frac{1}{3}\right)^9$$

(B) Probability of A winning the game at the end of 11th match is

$${}^{10}C_1\left(\frac{1}{2}\right)^2\left(\frac{1}{3}\right)^9 + {}^{10}C_2\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)^9$$

(C) If it is known that A wins the game at the end of 11th match then the probability that B will win only one match is $\frac{9}{11}$

(D) If it is known that A win the game at the end of 11th match then the probability that B will win none of the matches is $\frac{9}{13}$

For More Join: @IITJEE_Advanced

SECTION - III
(Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.
FOUR options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.
For each question, choose the option corresponding to the correct matching.
For each question, choose the option corresponding to the correct matching.
Full Marks : +3 If ONLY the option corresponding to the correct matching is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –1 In all other cases.

15. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(I)	If a, b, c are distinct natural randomly selected from first 100 natural numbers, then probability that a, b, c are individually divisible by both 2 and 3	(P)	$\frac{5}{28}$
(II)	Probability that in a year of 22nd century chosen at random there will be 53 Sundays	(Q)	$\frac{31}{32}$
(III)	If an unbiased coin is tossed 5 times then probability of getting at least 1 head is	(R)	$\frac{5}{12}$
(IV)	If $P(A \cup B)=\frac{3}{4}$, $P(A \cap B)=\frac{1}{4}$, $P(\bar{A})=\frac{2}{3}$, then $P(\bar{A} \cap B)$ is	(S)	$\frac{4}{1155}$
		(T)	$\frac{11}{25}$

(A) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ Q; IV $\rightarrow$ R (B) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ S

(C) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ R (D) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ Q; IV $\rightarrow$ S

16. A box contains 25 tickets, each with a different number from 1 to 25. Four tickets are drawn at random and without replacement. Let a, b, c, d be the numbers of the tickets drawn.

	List-I		List-II
(I)	The probability that min (a, b, c, d) < 10 is equal to	(P)	$\frac{189}{2530}$
(II)	The probability that the second smallest number chosen is 10 is equal to	(Q)	$\frac{286}{575}$
(III)	The probability that the sum of a + b + c + d is odd is equal to	(R)	$\frac{207}{230}$
(IV)	The probability that the product of abcd is even is equal to	(S)	$\frac{217}{230}$
		(T)	$\frac{1083}{1265}$

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(A) I → P; II → T; III → R; IV → S (B) I →T; II → P; III → Q; IV →S

(C) I → T; II → S; III → Q; IV → R (D) I →P; II → T; III → R; IV →Q

17. Three distinct numbers selected from {1, 2, ...6} the probability that

	List-I		List-II
(I)	They are in A.P. is	(P)	$\frac{1}{20}$
(II)	They are in G.P. is	(Q)	$\frac{3}{10}$
(III)	They are in H.P. is	(R)	$\frac{1}{10}$
(IV)	They are the sides of a triangle is	(S)	$\frac{7}{20}$
		(T)	$\frac{6}{35}$

(A) I → P; II → Q; III → S; IV → R (B) I → Q; II → P; III → R; IV → S

(C) I → Q; II → P; III → S; IV → R (D) I → T; II → Q; III → S; IV → R

18. Let $(1 - x + x^2)^{30} = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots\ldots\ldots + a_{60}x^{60}$.

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(I)	$\left(\frac{a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots\ldots}{a_{30}}\right)$	(P)	0
(II)	$4\left(\frac{a_0 + a_2 + a_4 + \ldots\ldots}{(3^{30}+1)}\right)$	(Q)	1
(III)	$6\left(\frac{a_1 + a_3 + a_5 + \ldots\ldots}{(1-3^{30})}\right)$	(R)	2
(IV)	$a_0a_1 - a_1a_2 + a_2a_3 - \ldots\ldots\ldots$	(S)	3
		(T)	4

(A) I → Q; II → R; III → S; IV → P (B) I → P; II → R; III → S; IV → Q

(C) I → Q; II→ S; III → R; IV → P (D) I → S; II → R; III → Q; IV → P

For More Join: @IITJEE_Advanced

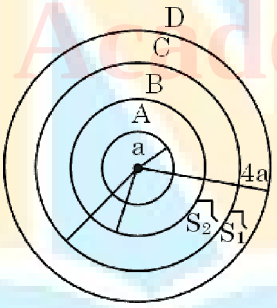
PHYSICS **Max. Marks: 60**

SECTION - I
(Maximum Marks : 24)

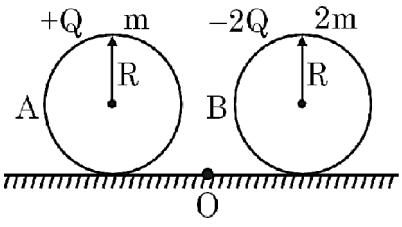
This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

19. An artificial satellite is moving in a circular orbit around the earth with a speed equal to $\frac{3}{8}$ times of the magnitude of escape velocity from the earth. If the satellite is stopped suddenly in its orbit and allowed to fall freely onto the earth, then the speed with which it hits the surface of the earth is $\sqrt{\frac{23}{n}}$ Km / s . Find n. (Take g at surface = 10 m/s^2 and R_e = 6400 Km)

20. In the figure A, B, C and D are four concentric spheres of radius a, 2a, 3a and 4a and contain, charges +3q, +6q, –9q and +12q respectively. In the beginning switches S_1 and S_2 are open. After earthing the sphere C Δq_1 amount of charge flows from the sphere C to the earth. If the sphere B is also earthed, then Δq_2 amount of charge flows from the sphere B to the earth, find the value of $\left|\frac{\Delta q_1}{\Delta q_2}\right|$ (switches are connected to earth)

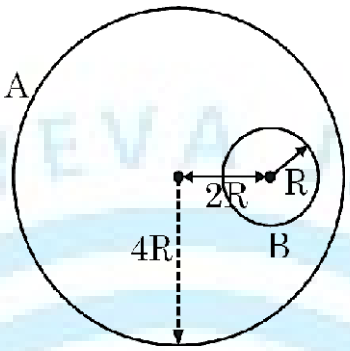


21. A hollow non conducting sphere A and a solid non conducting sphere B of equal radius 'R' and masses m and 2 m are kept at a large distance apart on a rough horizontal surface. Charge on the two spheres A and B are Q and –2Q respectively. Charges are distributed uniformly and remain constant and uniform as the spheres come closer. Friction is sufficient to support pure rolling and the kinetic energy of the two spheres just before collision is K_A and K_B.
Find $100\times\frac{K_A}{K_B}$.

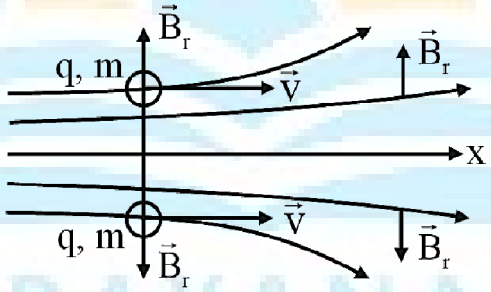


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22. Figure shows a uniformly charged spherical shell B of charge ($q_B = q$) kept inside uniformly charged spherical shell A of charge ($q_A = q$). Let electric field due to A and B at any point are $\vec{E}_A$ and $\vec{E}_B$ respectively. If dV represents elementary volume, then the value of $\frac{\int \text{entire space } \vec{E}_B.\vec{E}_B dV}{\int \text{entire space } \vec{E}_A.\vec{E}_B dV}$ will be

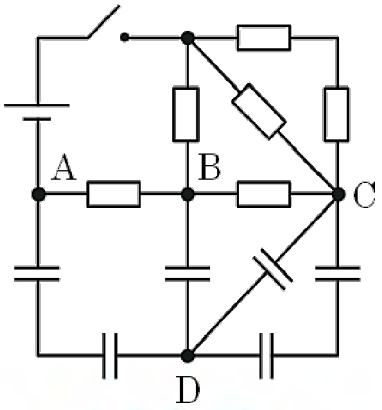


23. In an axially symmetric magnetic field, there is a dumbbell - a light non-conducting rod with small charged spheres at its ends. The masses and charges of the spheres are equal and equal to m and q. The dumbbell is perpendicular to the axis of symmetry (the x-axis), and its center is located on this axis (see figure). The projection of the magnetic field onto the radial direction (perpendicular to the axis) at a distance equal to the radius of the dumbbell is the same everywhere and equal to B_r. The axial component of the field varies along the axis. At the moment of time t_0, the dumbbell is given a velocity v along the x-axis. Neglect gravity. To what maximum distance L_{max} (in m) from the initial position does the center of the dumbbell move away? Take: m = 1 kg, v = 1 m/s, B_r = 0.5 T, q = 1C.



24. A circuit is made out of a battery, a switch, resistors and capacitors as shown in the figure.
The resistors all have a resistance of R, the capacitors all have a capacitance of C and the battery has a voltage of U. The point A is connected to the ground. In the beginning the switch is open and all the capacitors have no charge. The switch is closed at t = 0. The potential at point D in steady condition is αU. The value of α is.

For More Join: @IITJEE_Advanced



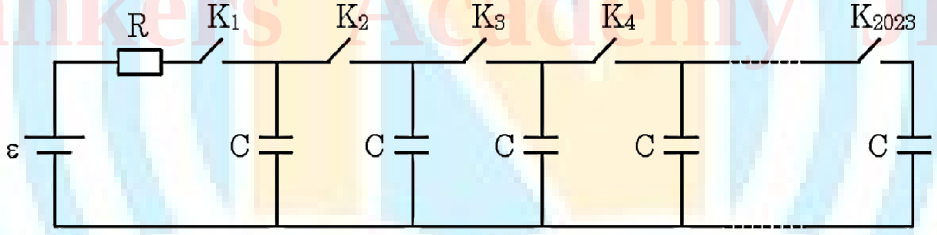
25. In two identical circuits, the diagrams of which are shown in the figure, at the initial moment all capacitors are uncharged and all switches are open. In these circuits, the switches are closed sequentially:

• In the first sequence – from switch K_1 to switch K_{2023} ($K_1 \rightarrow K_{2023}$),

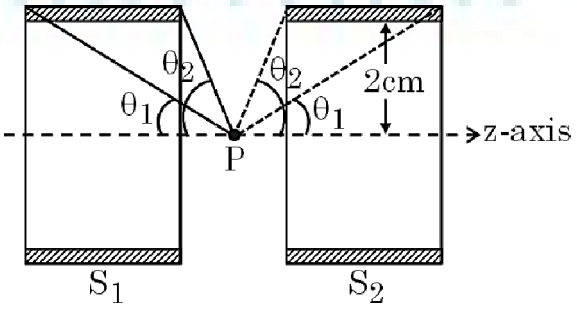
• In the second sequence – in the reverse order ($K_{2023} \rightarrow K_1$),

waiting each time long enough for the capacitors to fully recharge.

The ratio of heat loss in R in last step of sequence-2 to the heat loss in R in last step of sequence-1 is α. The internal resistance of the source can be neglected. The value of $\sqrt{\alpha}$ is



26. Two solenoids, each of radius 2 cm are placed coaxially. Determine the magnetic field (in μT) at the centre of the gap if a current of 2 A flows through the two solenoids in series in same direction. Assume that each solenoid contains $\left(\frac{40}{\pi}\right)$ turns per cm. $\theta_1 = 37°$, $\theta_2 = 53°$.



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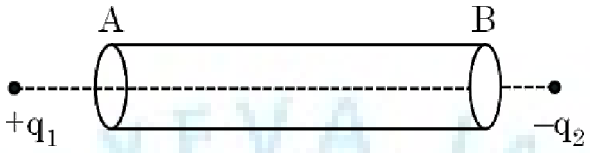
SECTION – II

(Maximum Marks : 24)

This section contains **SIX(06)** multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE THAN ONE** option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

27. A thin conducting rod AB is introduced in between the two charges $+q_1$ and $-q_2$ as shown in figure. For this situation mark the correct statement (s).



(A) The total force experienced by q_1 is vector sum of electric force due to $-q_2$ and due to induced charges on rod.

(B) The end A will become negatively charged.

(C) The total magnitude of force acting on $+q_1$ will be greater than as compared to the case without rod.

(D) The total magnitude of force acting on $-q_2$ will be greater than as compared to the case without rod.

28. In figure A, a stationary spacecraft of mass M is passed by asteroid A of mass m, asteroid B of the same mass m, and asteroid C of mass 2 m. The asteroids move along the indicated straight paths at the same speed; the perpendicular distances between the spacecraft and the paths are given as multiples of R. Figure B gives the gravitational potential energy U(t) of the spacecraft-asteroid system during the passage of each asteroid treating time t = 0 as the moment when separation is minimum. Which asteroid corresponds to which plot of U(t)?

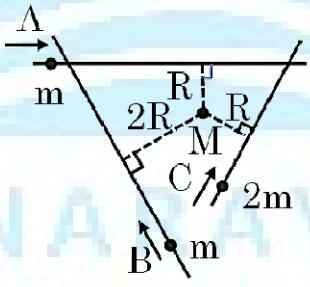


Figure A

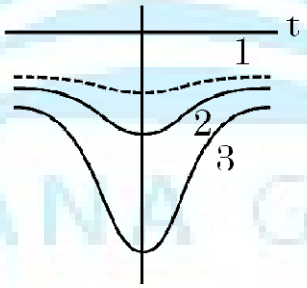


Figure B

(A) B – 1 (B) A – 2 (C) C – 3 (D) A – 3

29. Two very long, parallel conductors of negligible cross-section are placed at a distance of 2d apart, each carrying a current of the same magnitude I. Let us consider the x-axis as symmetry of the two conductors lying in the plane perpendicular to the plane of the conductors. Determine, on the x-axis for $0 \le x \le 4d$.

For More Join: @IITJEE_Advanced

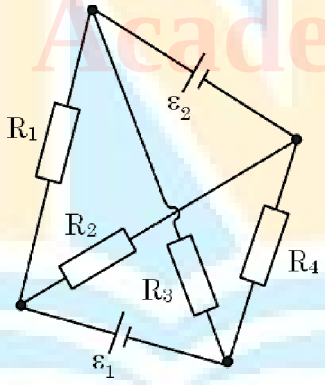
(A) The maximum magnitude of magnetic field if both wires are carrying current in the same direction is $\frac{\mu_0 I}{2\pi d}$

(B) The minimum magnitude of magnetic field if both wires are carrying current in the same direction is zero.

(C) The maximum magnitude of magnetic field if both wires are carrying current in the opposite direction is $\frac{\mu_0 I}{2\pi d}$

(D) The minimum magnitude of magnetic field if both wires are carrying current in the opposite direction is $\frac{\mu_0 I}{17\pi d}$

30. An electrical circuit is a tetrahedron, with two opposite edges containing the first and second ideal DC voltage sources with EMFs ε_1 and ε_2, respectively. The other edges contain four resistors R_1, R_2, R_3 and R_4 (see figure). The values of the EMFs of the sources and the resistance of one of the resistors are known: $\varepsilon_1 = 6V$, $\varepsilon_2 = 12V$, $R_2 = 3\Omega$. The current through the first source is $I_1 = 1A$ and does not change when the polarity of this source is reversed. The current through the second source, with the polarity indicated in the figure, is $I_2 = 2A$.



(A) The value of R_1 is 6 Ω

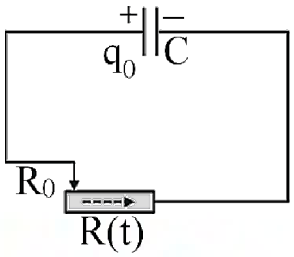
(B) The value of R_2 is 12 Ω

(C) The value of current through the second source if first source is removed from the circuit is 2A.

(D) The value of current through the second source if the first source is short circuited is 4A.

For More Join: @IITJEE_Advanced

31. To optimize the discharge process, a charged capacitor of capacitance C is connected to a rheostat, whose resistance can be varied from the maximum value R_0 down to zero. The initial charge of the capacitor is q_0.



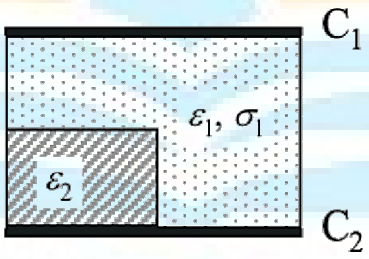
(A) The variation of R as a function of time t so that current remains constant during the discharge of the capacitor is $R_0\left(1-\frac{t}{R_0C}\right)$

(B) The time T in which the capacitor completely discharges is R_0C.

(C) The time T in which the capacitor completely discharges is $2R_0C$

(D) Total heat released in the resistor is $\frac{q_0^2}{4c}$

32. Two large parallel square conductor plates C_1 and C_2 of side length a are separated by two types of media. The cross section is shown in the figure. The distance between the plates is d. Medium-2 is a rectangular insulator slab with dielectric constant ε_2 thickness $\frac{d}{2}$ and side lengths a and $\frac{a}{2}$, respectively. The rest of the space between the plates is filled with
medium-1 with conductivity σ_1 and dielectric constant ε_1.



(A) The resistance between the two plates is $\frac{2d}{\sigma_1 a^2}$

(B) The resistance between the two plates is $\frac{d}{\sigma_1 a^2}$

(C) The capacitance between the two plates is $\frac{\epsilon_0 a^2}{d}\left(\epsilon_1+\frac{\epsilon_1\epsilon_2}{2(\epsilon_1+\epsilon_2)}\right)$

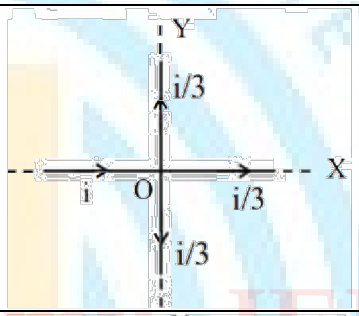
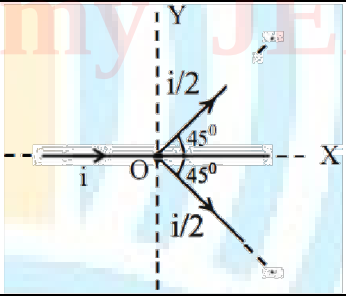
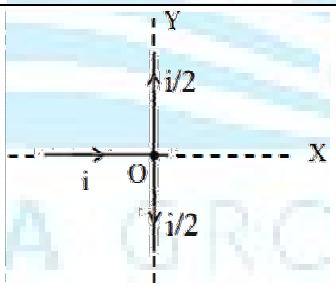
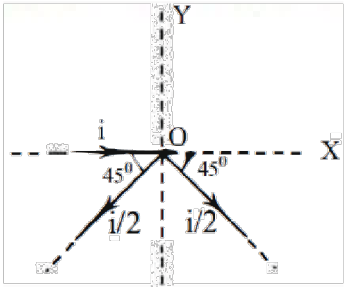
(D) The capacitance between the two plates is $\frac{\epsilon_0 a^2}{d}\left(\frac{\epsilon_1}{2}+\frac{\epsilon_1\epsilon_2}{\epsilon_1+\epsilon_2}\right)$

For More Join: @IITJEE_Advanced

SECTION - III
(Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.
FOUR options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.
For each question, choose the option corresponding to the correct matching.
For each question, choose the option corresponding to the correct matching.
Full Marks : +3 If ONLY the option corresponding to the correct matching is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –1 In all other cases.

33. List-II gives four situations in which three (in 2, 3, 4) and four (in 1) semi-infinite current carrying wires are placed in XY-plane as shown. The magnitude and direction of current is shown in each figure. List-I gives statements regarding the x and y components of magnetic field at a point P whose coordinates are P (0, 0, d). Match the statements in list-I with the corresponding figures in list-II.

	List-I		**List-II**
(I)	The x component of magnetic field at point P is zero	(P)	
(II)	The z component of magnetic field at point P is zero	(Q)	
(III)	The magnetic of magnetic field at point P is $\frac{\mu_0 i}{4\pi d}$	(R)	
(IV)	The magnitude of magnetic field at point P is less than $\frac{\mu_0 i}{2\pi d}$	(S)	
		(T)	None of these

For More Join: @IITJEE_Advanced

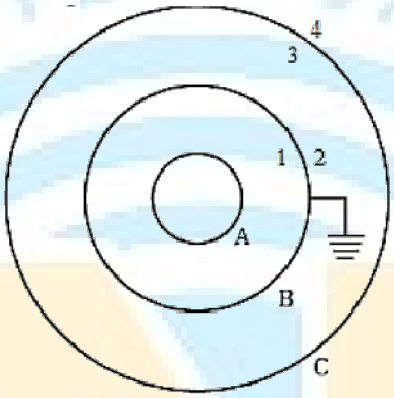
(A) I → P, Q, S; II → P, Q, R, S; III → T; IV → P, Q, R, S

(B) II → P, Q, R; II → P, Q, R, S; III → R; IV → P, Q, R, S

(C) III → P, Q, S; II → P, Q, S; III → P, R; IV → P, Q, R, S

(D) IV → P, Q, R; II → P, Q, R, S; III → R, S; IV → P, R, S

34. Figure shows three concentric thin spherical shells A, B and C of radii R, 2R, and 3R, respectively. The shell B is earthed and A and C are given charges q and 2q, respectively. If the charge on surfaces 1, 2, 3 and 4 are q_1, q_2, q_3, and q_4 respectively, then match the following list



	List-I		List-II
(I)	q_1	(P)	$\frac{2}{3}q$
(II)	q_2	(Q)	$\frac{4}{3}q$
(III)	q_3	(R)	$-\frac{4}{3}q$
(IV)	q_4	(S)	$-q$
		(T)	$-\frac{2q}{3}$

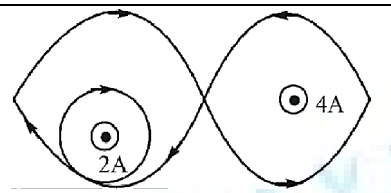
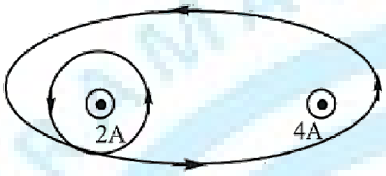
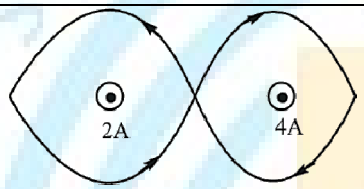
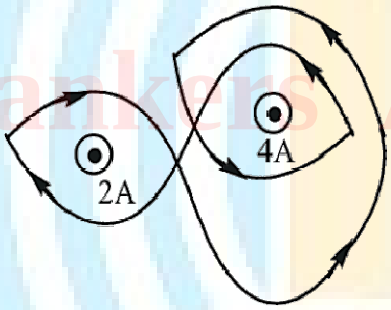
(A) I → P; II → Q; III → S; IV → R (B) I → Q; II → P; III → S; IV → R

(C) I → S; II → R; III → Q; IV → P (D) I → R; II → S; III → P; IV → Q

For More Join: @IITJEE_Advanced

35. A current carrying wire carries a current of 2A, lies along the z-axis which is out of page, another wire carrying current of 4A in same direction lies parallel to the first.

Match the value of $\frac{1}{\mu}\int \vec{B} \cdot \overrightarrow{d\ell}$ for the loops in List-I, with their values in List-II

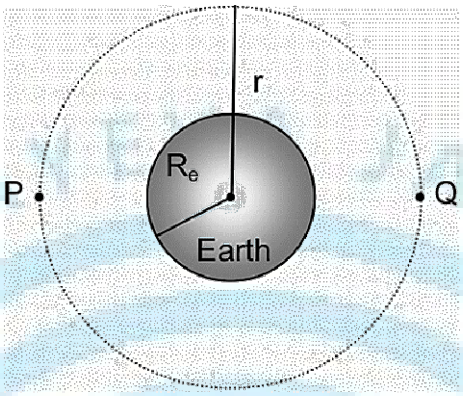
	List-I		List-II
(I)		(P)	0
(II)		(Q)	–2
(III)		(R)	6
(IV)		(S)	8
		(T)	2

(A) I → P; II → S; III → Q; IV → R (B) I → Q; II → P; III → R; IV → Q

(C) I → Q; II → R; III → P; IV → S (D) I → P; II → S; III → T; IV → R

For More Join: @IITJEE_Advanced

36. Two satellites 'P' and 'Q' are moving in a circular orbit about the Earth. The masses of both satellites are same and equal to 500 kg. The radius of circular orbit is 9600 km. Satellite 'P' is moving in clockwise direction and Q anticlockwise. Both satellites undergo a collision which is perfectly inelastic. Assuming earth is at rest. Given mass of earth = 6×10^{24} kg, radius of earth = 6400 km and gravitational constant G = 6.67×10^{-11} N-m^2/kg^2.



Match each entry in List-I with an appropriate value from List-II and choose the correct option.

	List-I		List-II
(I)	Time period of satellite before collision in sec	(P)	30.99×10^{12}
(II)	Magnitude of total energy of system before collision in joule	(Q)	15.68×10^{6}
(III)	Angular momentum of each satellite before collision in kg-m^2/sec	(R)	20.84×10^{9}
(IV)	Magnitude of total energy of system just after collision.	(S)	9.34×10^{3}
		(T)	41.68×10^{9}

The correct option is

(A) I → P; II → R; III → T; IV → S (B) I → Q; II → R; III → S; IV → P

(C) I → S; II → R; III → P; IV → T (D) I → Q; II → P; III → T; IV → S

For More Join: @IITJEE_Advanced

CHEMISTRY **Max. Marks: 60**

SECTION - I
(Maximum Marks : 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

37. H_2 gas desorbs from two metal surfaces A and B by first-order kinetic. The activation energy of desorption from surfaces A and B is 15 Kcal mol^{-1} and 25 Kcal mol^{-1} respectively.

The pre-exponential factors are 2×10^{13} sec^{-1} and 5×10^{11} sec^{-1}, respectively at 500K.

The ratio of surface courage remaining after 10 sec. is $e^{x \times 10^7}$. The value of x _____.

(Use: $e^{-15.10} = 2.75 \times 10^{-7}$, $e^{-25.17} = 1.17 \times 10^{-11}$)

38. 400 mL of 1.5% (w/w) KCl solution of density 1.10 g mL^{-1} is just sufficient to coagulate 250 mL of $Fe(OH)_3$ sol. The coagulation value of KCl in mmol L^{-1} is _____.

39. $CoCl_2 + 7KNO_2 + 2CH_3COOH \; K_3\left[Co\left(NO_2\right)_6\right] + 2KCl + 2CH_3COOK + NO + H_2O$

$K_3[Co(NO_2)_6]$ absorbs 360 nm light and pairing energy is 250 KJ/mole

(Take planck's constant (h) = 6.0×10^{-34} J-s, speed of light(c) = 3.0×10^8 m/s, Avogadro number 6.0×10^{23} mol^{-1})

The magnitude crystal field stabilization energy (CFSE) _____ KJ mol^{-1}.

40. For the complex $[Co(H_2O)_2(F)_2ClBr]$

Number of geometrical isomers = X

Number of optically active isomer = Y

Then the value of $\frac{Y}{X}$ is _____.

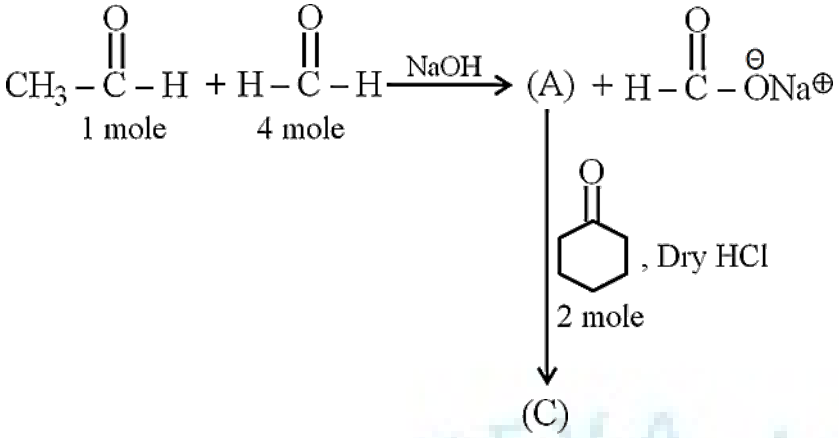
41. $CuCl_2$ + light yellow complex compound (aq) $\longrightarrow \underset{ppt}{\downarrow A} + KCl$ (unbalanced)

The light-yellow complex has No geometrical and optical isomer and can show ligand isomerism.

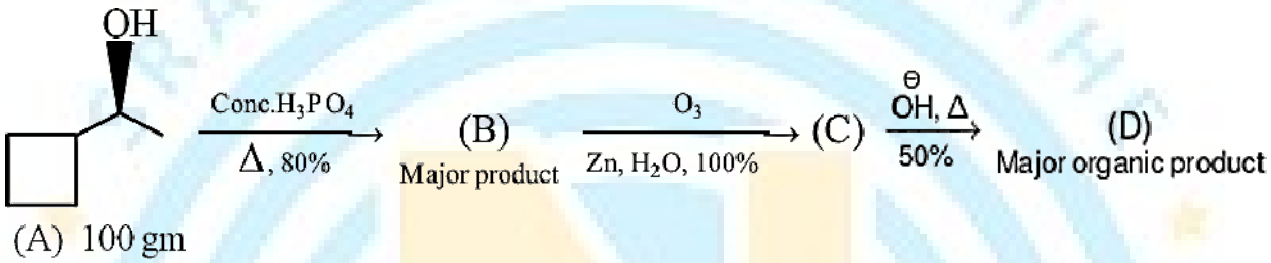
A is brown precipitate and co-ordination compound.

Its spin only magnetic moment is X B.M. and then value of (X + 5.50) would be _____.

For More Join: @IITJEE_Advanced

42. 

Number of sp^3 carbons in product C is

43. 

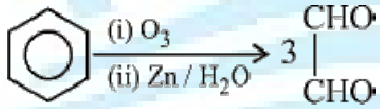
Respective % yields are given with each reaction

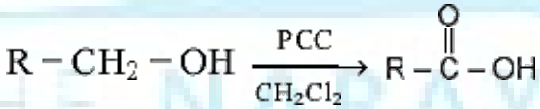
X gm of product D is formed

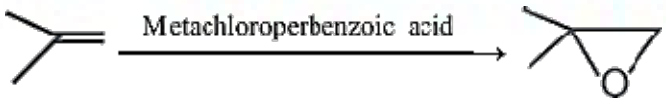
The value of 10X is

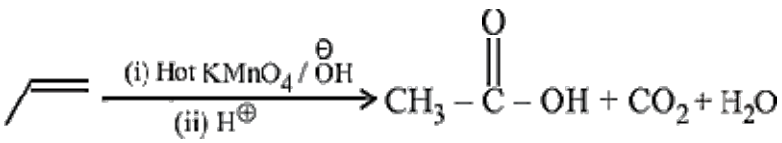
(Molecular weight of C = 12, O = 16, H = 1]

44. How many of the following are correctly matched with their product?

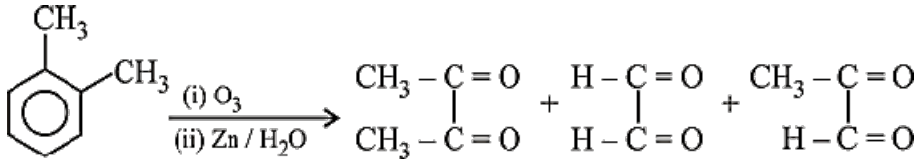
(i) 

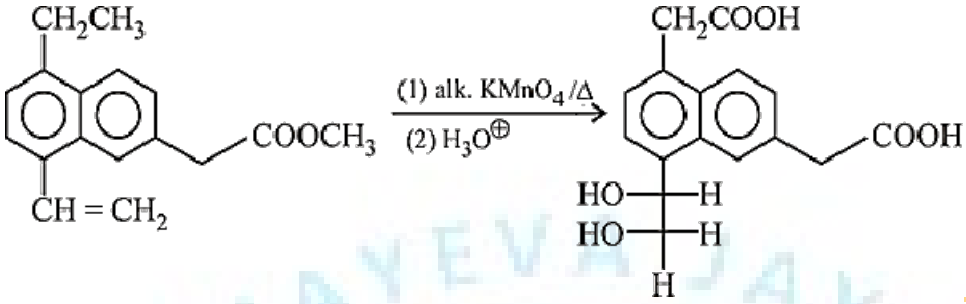
(ii) 

(iii) 

(iv) 

For More Join: @IITJEE_Advanced

(v) 

(vi) 

SECTION – II
(Maximum Marks : 24)

This section contains **SIX(06)** multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE THAN ONE** option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

45. Consider the given statements about critical micelle concentration in aqueous solution and select the correct statement.

(A) The CMC increases as the length of the hydrophobic tail increases

(B) Ionic surfactants have higher CMC as compared to non-ionic surfactants containing equivalent hydrophobic groups

(C) Non-ionic surfactants have higher CMC as compared to ionic surfactants containing equivalent hydrophobic groups

(D) Surfactants containing more than one hydrophilic groups show smaller CMC than those with one hydrophilic group with the hydrophobic group being equivalent in both

46. Consider following complex

(P) $[Ni(PPh_3)_2Cl_2]$ (Q) $K_3[Co(CO_3)_3]$

(R) $K[PtCl_3(C_2H_4)]$ (S) $[Cr(en)_2Cl_2]^{\oplus}$

Which statement(s) is/are correct regarding given complexes

(A) Complex P shows temperature dependent magnetic behaviour

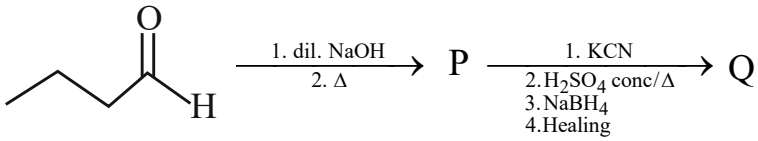
(B) Only P, R and S can show geometrical isomerism

(C) Complex R has C-C bond stronger than free ethene $H_2C{=}CH_2$

(D) Cis-geometrical isomer is optically active isomer for R but not for P

For More Join: @IITJEE_Advanced

47. Considering the reaction sequence given below, the correct statement(s) is/are



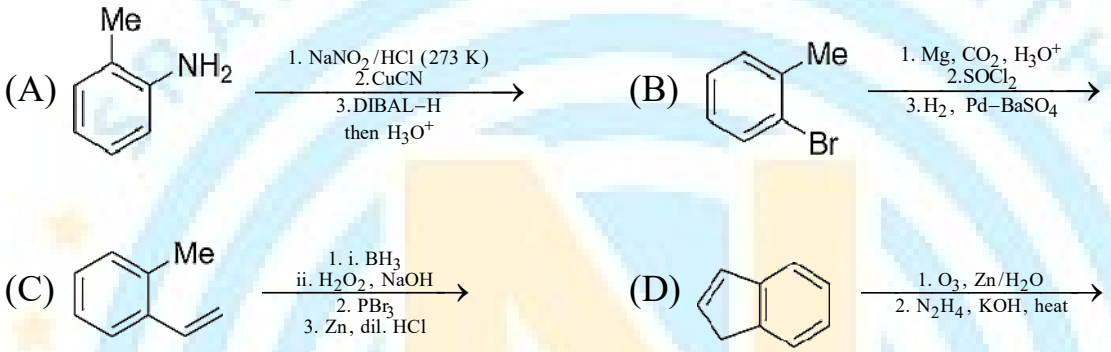
(A) P can show tautomerism

(B) Treating P with LiAlH4 followed by acidification give unsaturated alcohol

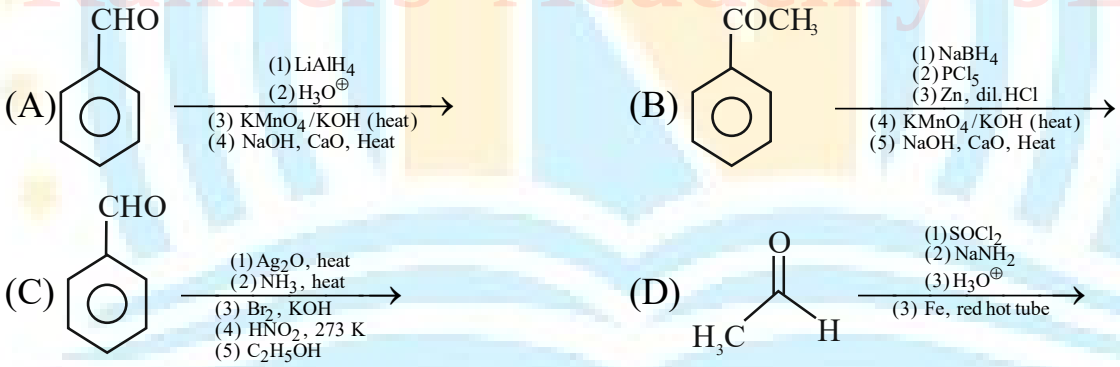
(C) Treating Q with a Zn-Hg | HCl gives cyclic ether

(D) Q on reduction with DIBAI-H/Toluene at −76°C give an optically active compound

48. The reaction sequence(s) that would lead to 2-methyl benzaldehyde is/are



49. The reaction sequence(s) that would lead to benzene as the major product is/are



50. For each of the following coordination compounds, state whether the given description about its magnetic nature and type of isomerism is correct.

Select the correct option(s).

(A) Tetraamminedichloridochromium(III) chloride is paramagnetic and exhibits cis-trans isomerism.

(B) Pentaaquachloridotitanium(III) nitrate is paramagnetic and exhibits ionisation isomerism.

(C) Amminechlorido(ethylenediamine)platinum(II) nitrate is diamagnetic and exhibits ionisation isomerism.

(D) Tetraamminedinitratocobalt(III) nitrate (high-spin) is paramagnetic and exhibits cis-trans isomerism

For More Join: @IITJEE_Advanced

SECTION - III
(Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.
FOUR options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.
For each question, choose the option corresponding to the correct matching.
For each question, choose the option corresponding to the correct matching.
Full Marks : +3 If ONLY the option corresponding to the correct matching is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –1 In all other cases.

51. LIST-I contains metal species and LIST-II contains their properties.

	LIST-I		**LIST-II**
(I)	$[Cr(CN)_6]^{4-}$	(P)	t_{2g} orbitals contain 4 electrons
(II)	$[RuCl_6]^{2-}$	(Q)	μ(spin-only) = 4.9 BM
(III)	$[Cr(H_2O)_6]^{2+}$	(R)	low spin complex ion
(IV)	$[Fe(H_2O)_6]^{2+}$	(S)	metal ion in 4+ oxidation state
		(T)	d^4 species

[Given: Atomic number of Cr = 24, Ru = 44, Fe = 26]

Match each metal species in LIST-I with their properties in LIST-II, and choose the correct option

(A) I $\rightarrow$ R, T; II $\rightarrow$ P, S; III $\rightarrow$ Q, T; IV $\rightarrow$ P, Q

(B) I $\rightarrow$ R, S; II $\rightarrow$ P, T; III $\rightarrow$ P, Q; IV $\rightarrow$ Q, T

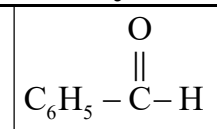
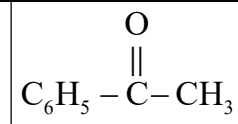
(C) I $\rightarrow$ P, R; II $\rightarrow$ R, S; III $\rightarrow$ R, T; IV $\rightarrow$ P, T

(D) I $\rightarrow$ Q, T; II $\rightarrow$ S, T; III $\rightarrow$ P, T; IV $\rightarrow$ Q, R

52. Match List-I with List-II

	LIST-I		**LIST-II**
(I)	$CH_3-\overset{\overset{O}{\|}}{C}-H$	(P)	can be oxidised by Tollen's reagent
(II)	$CH_3-\overset{\overset{O}{\|}}{C}-CH_3$	(Q)	can be oxidised by Felling reagent

For More Join: @IITJEE_Advanced

(III)		(R)	Undergoes Cannizzaro reaction with NaOH
(IV)		(S)	Undergo aldol condensation with NaOH, Heat
		(T)	Water insoluble

Match each metal species in List-I with their properties in List-II, and choose the correct option

(A) I → P, Q, S, II → S; III → P, R, T; IV→ S, T

(B) I → Q, S; II → S, T; III → P, S, T; IV → R, S

(C) I → S, T; II → R; III → P, Q, S, IV → P, Q

(D) I → Q, R, S; II → R, T; III → S, T; IV → P, S, T

53. List-I contains metal species and List-II contains their products

	LIST-I		**LIST-II**
(I)	$[CoF_6]^{3-}$	(P)	t_{2g} orbitals contain 6 electrons
(II)	$[Co(H_2O)_6]^{3+}$	(Q)	μ(spin-only) = zero BM
(III)	$[CoCl_4]^{2-}$	(R)	Low spin complex ion
(IV)	$[Zn(CN)_4]^{2-}$	(S)	sp^3d^2
		(T)	sp^3

Match each metal species in List-I with their properties in List-II, and choose the correct option

(A) I → R; II → R, Q, S; III → R, S; IV → P, S

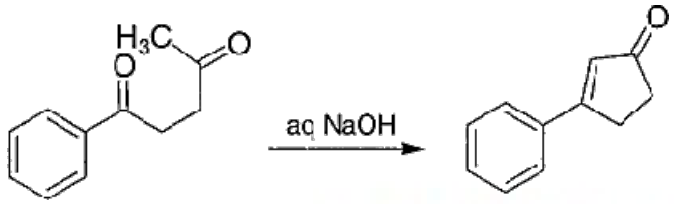
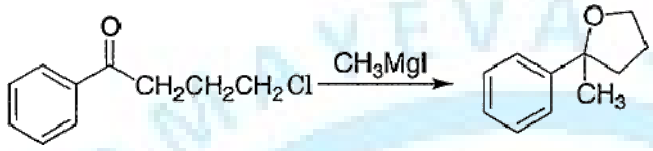
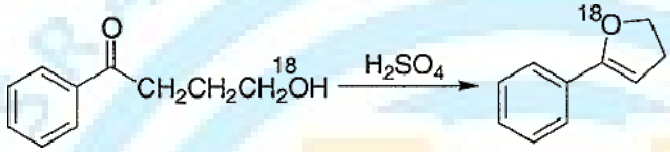
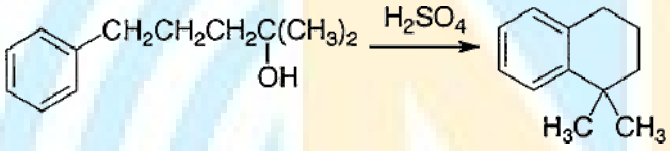
(B) I → P; II → P, S; III → P, Q; IV→ P, Q, R

(C) I → R; II → Q, R, S; III → P, Q; IV → Q, R, S

(D) I → S; II → P, Q, R; III → T; IV→ Q, T

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54. Match the reactions in List-I with appropriate types of steps/reactive intermediate involved in these reactions as given in List-II

	LIST-I		LIST-II
(I)		(P)	Nucleophilic substitution
(II)		(Q)	Electrophilic substitution
(III)		(R)	Dehydration
(IV)		(S)	Nucleophilic addition
		(T)	Carbanion

Match each metal species in List-I with their properties in List-II, and choose the correct option

(A) I → Q, R, S; II → P, T; III → Q, S; IV→ P, Q

(B) I → R, S, T; II → P, S; III → R, S; IV → Q, R

(C) I → Q, S; II → P, Q; III → R, T; IV → P, Q

(D) I → S, T; II → P, Q; III → P, R, T; IV → S, T

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